README demo

This file contains a description of the code `demo.m`. The aim of this code is to demonstrate the use of the functions `ACFZcrit.m` and `DDR.m`. Run the code to obtain the estimates of the number of shocks in a synthetic dataset using all the frequencies or frequency bands. The code consists of 4 parts:

Part A

This part of the code generates a $T \times N$ panel data as a realization of double sequence of random variables $\{X_{it}, i \in \mathbb{N}, t \in \mathbb{Z}\}$ which admits the representation

$$X_{it} = \Lambda_{i1}(L)u_{1t} + \cdots + \Lambda_{iq}(L)u_{qt} + e_{it}$$

where $\Lambda_{ij}(L) = \sum_{k=0}^{\infty} \Lambda_{ij,k}L^k$, L denotes the lag operator, u_{jt}, $j = 1, \dots, q$ are the shocks and e_{it} are the idiosyncratic components.

The filters $\Lambda_{ij}(L)$ are randomly generated (independently from u_{jt}'s and e_{it}'s) as

$$\Lambda_{ij}(L) = b_{ij}^{(0)} \left(1 - b_{ij}^{(1)} L\right)^{-1} \left(1 - b_{ij}^{(2)} L\right)^{-1}, \qquad i = 1, \dots, N; \quad j = 1, \dots, q$$

with i.i.d. mutually independent coefficients $b_{ij}^{(0)} \sim \mathcal{N}(0,1)$, $b_{ij}^{(k)} \sim \mathcal{U}[-0.8, 0.8]$, $k = 1, 2$, where $\mathcal{N}(0,1)$ and $\mathcal{U}[a,b]$ denote the standard normal distribution and the uniform distribution with support $[a,b]$, respectively. The shocks u_{jt} are i.i.d. $\mathcal{N}(0,1)$ and mutually independent.

The common components $\sum_{j=1}^{q} \Lambda_{ij}(L)u_{jt}$ are normalized to a have (population) variance equal to one. The idiosyncratic components e_{it} are generated as realizations of i.i.d. standard normal random variables independent of all other variables.

The default values are: `T=300`, `N=300` and `q=2`.

Part B

This part of the code estimates the number of shock q by using the function

```
[kDER, kDGR, kDDR] = ACFZcrit(X, qmax,c)
```

All the frequencies are considered.

Inputs

- `X` is the dataset generated in Part A.

- `qmax` is a predetermined upper value to the number of shocks.

  The code will return an error if `qmax`$>\min(N, 2M_T+1)$ because the auxiliary function `DynamicEigenvaluesPERS.m` computes the $\min(N, 2M_T+1)$ non-zero eigenvalues of the spectral density matrix (see equation (2.5) in the paper) and the criteria require up to `qmax+2` eigenvalues to be computed.

- `c` is a positive and bounded scalar value used to compute the bandwidth M_T (see Section 2.2. in the paper) as $M_T = \lfloor c\sqrt{T} \rfloor$. If the input is omitted, the default value `c=0.75` is used.

Outputs

- `kDER` is the estimated value of `q` obtained by using the estimator $\widehat{k}_{DER}$ in equation (2.9) of the paper.

- `kDGR` is the estimated value of `q` obtained by using the estimator $\widehat{k}_{DGR}$ in equation (2.10) of the paper.

- `kDDR` is the estimated value of `q` obtained by using the estimator $\widehat{k}_{DDR}$ in equation (2.8) of the paper.

Please see the description in `ACFZcrit.m` for further details.

Part C

This part of the code estimates the number of shocks q by using the function

```
kDDR=DDR(X,qmax,c,freqband)
```

Inputs

- `X`, `qmax` and `c`: see Part B.

- `freqband` is the frequency band considered to compute the DDR criterion (See Section 2.2. in the paper). The code considers two examples, `freqband1=[0  2*pi/80]` (long-run frequencies) and `freqband2=[2*pi/32 2*pi/6]` (business-cycle frequencies).

Outputs

- `kDDR` is the estimated value of `q` obtained by using the estimator $\widehat{k}_{DDR}$ in equation (2.8) of the paper, where $[\omega_{\underline{\ell}}, \omega_{\overline{\ell}}]$ =`freqband`

Please see the description in `DDR.m` for further details.

Remark. The estimate `kDDR` obtained in Part B would be the same as the output of the function `DDR` if `freqband=[0, 2*pi]`. The function `ACFZcrit` could be modified to allow all the criteria to be computed considering all the frequencies or a given frequency band. However, a more complex code could be computationally costing for the simulation study.

Part D

Display the results.